



Hoang-Dung NGUYEN

Scientific Data Scientist | Machine Learning | Python | Predictive Modeling

Scientific Data Scientist with a PhD in Civil and Environmental Engineering and experience in predictive modeling, experimental data analysis and signal processing. Proficient in Python, R, SQL and Scikit-learn for preparing complex datasets, developing statistical models and evaluating their performance. Experienced in translating scientific questions into data-driven approaches and communicating technical results to multidisciplinary stakeholders.



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Orsay, France

PROJET DATA SCIENCE

Prediction of soil permeability from experimental and geophysical data

- Collected and preprocessed heterogeneous experimental datasets.
- Performed exploratory analysis and feature selection.
- Developed and compared regression models using Python and R.
- Evaluated model performance and interpreted the influence of key variables.
- Communicated results through scientific reports and presentations.

EXPÉRIENCES PROFESSIONNELLES

Actuarial Analyst Apprentice

ENTORIA

- Performed statistical analysis of health and protection insurance data and monitored client accounts.
- Built and analyzed client portfolios to monitor technical balance, claims ratios, and performance indicators.
- Contributed to pricing activities and the development of a 2027 pricing tool based on statistical data and actuarial assumptions.
- Extracted, cleaned, and processed data using SAS for technical and actuarial analysis; Developed reporting tools and management dashboards using Power BI and Power Query.

Ingénieur d'Études de Projets

Groupe Cochez Valenciennes

- Analyzed operational and safety data to identify risks, anomalies and trends.
- Developed reporting tools and dashboards for project performance monitoring.
- Collaborated with operational teams to translate field needs into analytical outputs.

Ingénieur de recherche

Université de Lille

- Designed experimental protocols and collected complex environmental and geophysical data from pilot sites.
- Prepared and analyzed experimental datasets using Python, R and GIS tools.
- Developed regression models to predict soil hydraulic properties and permeability.
- Processed signals, selected relevant variables and evaluated model performance using statistical metrics.
- Interpreted and communicated scientific results to researchers and project partners.
- Contributed to scientific publications and presented research results at conferences.

FORMATIONS

DataScientest

- **Business Data Analyst** 2025 - 2026

Université de Lille - France

- **Doctorat en Génie Civil** 2018 - 2022
- **Master 2 en Génie Urbain et Habitat** 2017 - 2018

COMPÉTENCES

Scientific Data Science

Exploratory data analysis, experimental design, statistical modeling, regression, predictive modeling, model evaluation, signal processing.

Programming and Machine Learning

- **Python** : Pandas, NumPy, Scikit-learn, Matplotlib
- **Langages et traitement des données** : SQL, SAS, R, Excel, Power Query, VBA

Artificial Intelligence

Generative AI and SEO: foundational knowledge

Data Visualization

Matplotlib, Power BI, Excel

Scientific Communication

Scientific reporting, interpretation of statistical results, presentation and communication to technical and non-technical stakeholders.

Depuis
Sept.
2025

Sept.
2023
-
Mai
2024

Sept.
2018
-
Août
2023